

ISTITUTO  
DI BIORBOTICA



Sant'Anna  
Scuola Universitaria Superiore Pisa

---

## P.A.S.H. - Parallel Adaptive Stiffness Heel

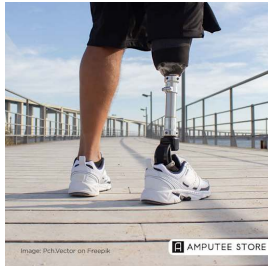
Andrea Diano · Andrea Goriani · Riccardo Mambrini

*Istituto di BioRobotica — Scuola Superiore Sant'Anna, Pisa*

24 aprile 2026

---

# Problem Identification & Context: The Limits of Passive Prostheses



- **Context:**

- Passive prostheses dominate the market for their reliability and low cost.
- They exhibit a **fixed dynamic response**.

- **Biomechanical Problem:**

- *Task-invariant response*: fixed stiffness cannot adapt between walking, running, or stair climbing.
- *Terrain-blind mechanics*: no impedance modulation.

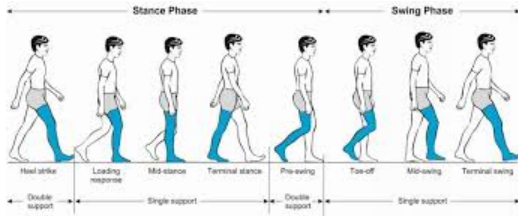
- **The Dilemma (Deformability vs. Energy):**

- *Soft heel*: absorbs impact but dissipates energy.
- *Rigid heel*: returns energy but damages the residual limb.

- **Project Goals:**

- **Zero Electronics**: No batteries, no sensors.
- **Low-Maintenance**: High durability and low cost.
- **Dynamic Cushioning**: Impact response optimization.

# Biomechanics of Locomotion and Gait Dynamics



## Froude Number ( $Fr$ )

$$Fr = \frac{v^2}{g \cdot L}$$

Critical threshold  $\approx 0.4$  for the walk-to-run transition.

### • Gait Cycle:

- Transition between *Stance Phase* (load absorption/push-off)
- *Swing Phase* (pendulum effect).

### • Speed Parameters:

- Governed by *Stride Length*
- *Step Frequency*. Increasing frequency distributes the load over time.

### • Focus: Heel Strike & Roll-over:

- Heel contact is the point of maximum stress at the **residual limb-socket interface**.
- Smooth roll-over is essential to minimize metabolic cost.

### • Strategic Choice:

- Cushioning localized at the heel.
- The forefoot remains rigid to ensure push-off (*Toe-off*).

# State of the Art: Passive Prosthetic Feet Overview



---

## SACH

**Pro:** Low cost, good initial shock absorption.

**Con:** No energy return, unnatural gait.

---

## ESR

**Pro:** Elastic energy return, smooth roll-over (carbon fiber).

**Con:** Fixed stiffness.

---

## Microprocessor (MPA)

**Pro:** Real-time adaptation, reduces tripping risk.

**Con:** High cost, weight, re-charging.

---

# Gap Analysis: The "Deformability-Energy Dilemma"

- **The Current Compromise**

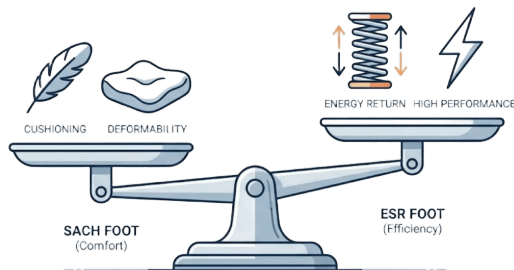
- *SACH*: good cushioning, but high energy dissipation.
- *ESR (Carbon Fiber)*: excellent return but fixed stiffness (harmful load transfer).

- **The Real Technological Gap:**

- Lack of a **100% passive** solution (no electronics) that is **dynamically adaptive**.

- **The Necessity:**

- Moving beyond the linearity of traditional materials to overcome mechanical limits.



# Innovative Proposal: Nature's Solution

## The Feline Model

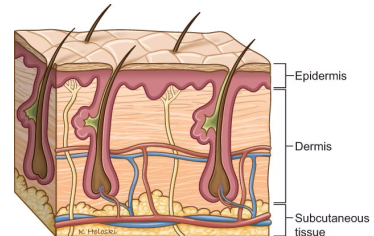
*The Paw Pad* is a masterpiece of non-linear biomechanics, capable of adapting to varying impact velocities.

## Anatomical Structure

- 1 **Epidermal:** Withstands intense ground interaction.
- 2 **Dermal:** Hydrostatic energy dissipation system.
- 3 **Subcutaneous:** Soft adipose tissue.

## Core Objective

Translating this exact morphology into a prosthetic heel.



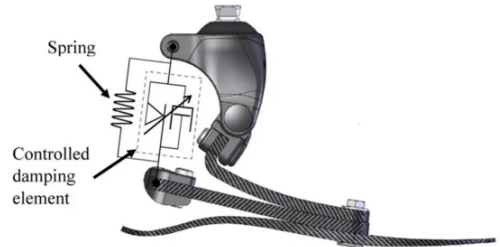
# Engineering Paradigm Shift

## P.A.S.H. Concept

### Parallel Adaptive-Stiffness Heel

#### Bioinspired Parallel Architecture:

- **Elastic Component:** manages metabolic efficiency during normal gait.
- **STF Damper:** "activates" only to absorb extreme impacts, without interfering at low speeds.



## The Result

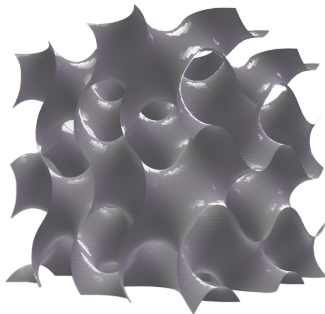
Perfect synergy between mechanical design and *Smart Materials*.

# Concept Development: Components and Materials



## 1. Elastic Component

**ESR Carbon Fiber Blade:** Acts as the main tendon. Stores propulsive energy and manages the *Toe-off* phase.



## 2. Structural Matrix

**Dermal Layer (TPMS):** Flexible TPU scaffold with gyroid structure. Sustains the baseline load and guides the spatial deformation.

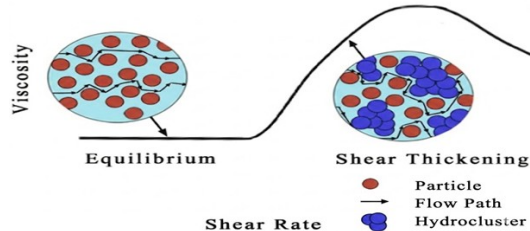


## 3. Intelligent Core

**Adipose Tissue (STF):** Confined non-Newtonian fluid (suspension of  $\text{SiO}_2$  nanoparticles in PEG). Modulates stiffness according to impact velocity.



# Concept Development: Smart Dynamic Response of STF



## Low Shear Rate (Walk, $Fr < 0.4$ )

- **STF State:** Low viscosity (Fluid).
- **Mechanics:** The TPMS matrix deforms elastically under the load.
- **Cushioning:** The heel "sinks" to absorb terrain irregularities.
- **Result:** Improved *roll-over*, damping without massively triggering the stiff blade.

## High Shear Rate (Run, $Fr > 0.4$ )

- **STF State:** Instant solidification.
- **Mechanics:** The damper locks its volumetric deformation.
- **Bridging:** The heel becomes a rigid bridge, dissipating all the energy.
- **Result:** Decreased damage from the high GRF, at the level of the limb-socket.

## • 1. Kinematic Response (The "Wall Effect" Risk)

- *Challenge:* Preventing premature STF hardening during oblique heel strikes at low speeds.
- *Solution:* **Rheological Tuning**. By precisely adjusting the  $\text{SiO}_2$  mass fraction, we calibrate the *critical shear rate* to ensure activation occurs only during high-energy tasks.

## • 2. Pendulum Effect (Distal Inertia)

- *Challenge:* Fluid mass increases the distal weight, raising metabolic cost during the *Swing Phase*.
- *Solution:* **Spatial Optimization**. STF is not distributed uniformly but encapsulated strictly within primary load-bearing lobes, concentrating the fluid in less peripheral areas.

## • 3. Material Fatigue & Leakage

- *Challenge:* Cyclic loading ( $10^6$  cycles) leading to TPU micro-fractures and fluid leakage.
- *Solution:* **Micro-compartmentalization**. The closed-cell TPMS structure ensures a *fail-safe* design. A localized puncture results in negligible performance degradation instead of catastrophic failure.

# Cost Analysis & Production

## 1. Bill of Materials (Unit Cost)

- **ESR Carbon Fiber:** ~100 € **TPU Scaffold:** ~10 € **STF Matrix (PEG + SiO<sub>2</sub>):** < 5 €

## 2. Production & Assembly

- **Prototyping:** SLS 3D Printing (TPU) + Nesting optimization.
- **Assembly:** Rapid mechanical sealing (<1h), highly standardized.

## 3. Market Position

- **Zero Electronics:** No need for sensors, batteries, or microcontrollers.
- **Low TCO (Total Cost of Ownership):** Passive design eliminates software calibration.
- **Target Cost:** ~500 € (P.A.S.H.) vs > 5000 € (Active Bionic Alternatives).

# Applications & Use Cases

## Transmetatarsal

*(Partial Foot Amputation)*

Passive heel compliance restores **push-off mechanics** at forefoot level. No electronics, no recalibration.

## Transtibial

*(Below-Knee Amputation)*

Variable stiffness tuning adapts to **gait speed** and user mass. Broadest clinical population.



## Transfemoral

*(Above-Knee Amputation)*

Passive heel shock absorption reduces **knee-joint loading** on the residual prosthetic knee unit.

## Smart Insoles

*(Runners / Astronauts)*

Adaptive heel cushioning for **intact limbs**: terrain-blind impact absorption in high-load or *microgravity* contexts.

- **Y. Xiao et al.**, "*A 3D-Printed Sole Design Bioinspired by Cat Paw Pad...*", *Polymers*, vol. 14, no. 16, p. 3270, 2022.
- **H. Tryggvason et al.**, "*Use of Dynamic FEA for Design Modification and Energy Analysis of a Variable Stiffness Prosthetic Foot,*" *Applied Sciences*, vol. 10, no. 2, p. 650, 2020.
- **J. Z. Laferrier et al.**, "*A Review of Commonly Used Prosthetic Feet for Developing Countries: A Call for Research and Development,*" *J. Nov. Physiother.*, vol. 8, no. 1, p. 380, 2018.

## Full Bibliography



Scan to access the full OneDrive project folder